

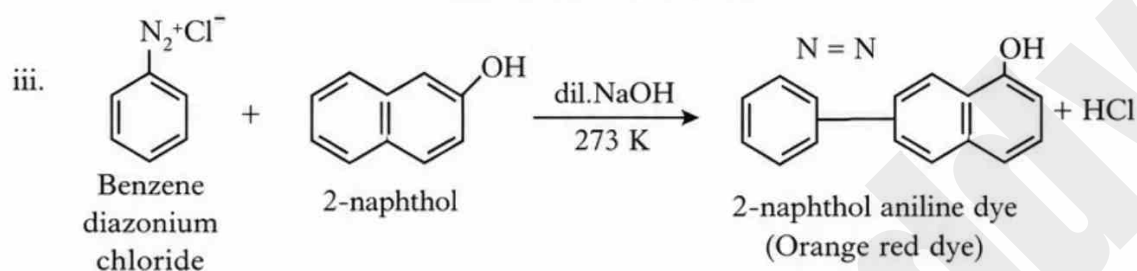
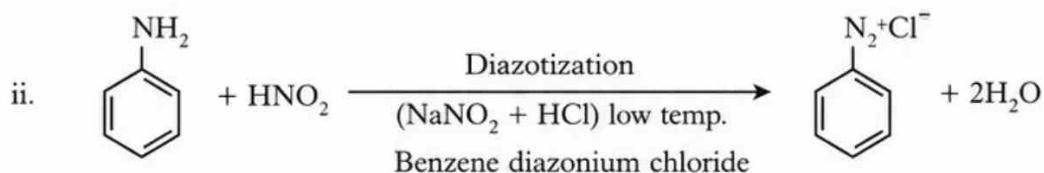
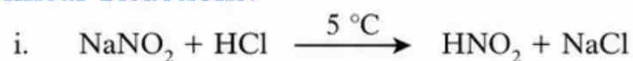
Experiment No. 15

Date: / /

Aim: To prepare 2-naphthol aniline dye from aniline.

Theory: It is an azo-coupling reaction. azo compounds contain $-N=N-$ group. First aniline undergoes diazotisation reaction with nitrous acid (HNO_2) at low temperature to form benzene diazonium chloride. It further undergo coupling with 2-naphthol to produce 2-naphthol aniline dye. 2-naphthol aniline dye (Benzene-azo-2-naphthol) is a scarlet red azo dye.

Chemical Reactions:



Apparatus: Beaker, conical flask, glass rod, ice bath, test tube, funnel, etc.

Chemicals: Aniline, sodium nitrite, conc. HCl, β naphthol, 10% NaOH.

Procedure:

1. Take 1 mL aniline in 2.5 mL of concentrated HCl in a 100 mL conical flask add 5 mL distilled water and cool the flask in an ice bath.
2. Take 1 g of $NaNO_2$ in another beaker add 5 mL distilled water to dissolve it. Cool this beaker in an ice bath.
3. Add ice cooled solution of $NaNO_2$ slowly to the cold solution of aniline and HCl with continuous shaking, to prepare benzene diazonium chloride.
4. Add 2 g of 2-naphthol in 15 mL of 10% NaOH in a test tube and dissolve it.
5. Now add benzene diazonium chloride solution to 2-naphthol in NaOH with constant stirring with glass rod.
6. Orange red dye crystals of 2-naphthol aniline dye is formed. Filter it in a buchner funnel. Wash this dye with water.
7. Dry it and find weight of crystals.

Observation and Result:

- i. Yield of 2-naphthol aniline dye crystals = 0.8 g
- ii. Colour of 2-naphthol aniline dye = orange red

Note: Azo-dye synthesis is nearly quantitative, so one should use amount of reactants as per the equations. Any excess of certain reactants may cause decomposition of unused material and may produce 'tar'.

Remark and sign of teacher:

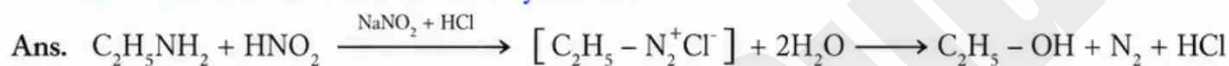
MCQ

Select the correct answer for the following multiple choice type questions.

- The colour of pure aniline is
 a. blue b. red ☒ c. faint yellow d. orange
- The colour of β -naphtholaniline dye is
☒ a. orange b. yellow c. red d. pink
- The temperature maintained throughout the above experiment should be
 a. more than 25°C ☒ b. less than 25°C c. less than 5°C d. 298 K
- Addition of NaNO_2 to the cold solution of aniline and HCl is reaction.
 a. exothermic b. endothermic c. substitution ☒ d. condensation
- Formation of benzenediazonium chloride is
 a. coupling reaction b. di azo-coupling reaction
☒ c. diazotization d. combination

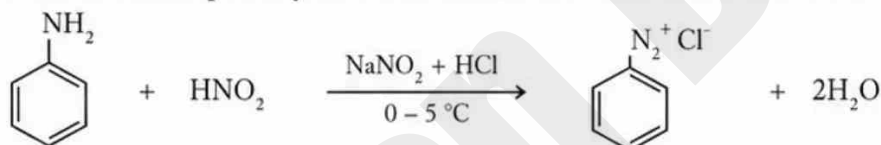
Short answer questions

1. Write diazotization reaction of ethylamine.



2. What is diazotization? Write the reaction involved during diazotization of aniline and nitrous acid.

Ans. The conversion of primary aromatic amines into diazonium salt is known as diazotization.



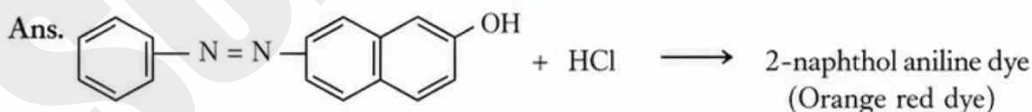
3. What do you understand by coupling reaction?

Ans. A diazonium salts react with certain aromatic compounds having general formula $\text{Ar} - \text{H}$ or $\text{Ar} - \text{OH}$ or $\text{Ar} - \text{NH}_2$ etc. to form azo-compounds. This reaction is called coupling.

4. Which of the following compound on diazotisation followed by coupling with 2 naphthol will form an azo-dye? (i. p-toluidine ii. benzylamine iii. N-methylamine)

Ans. ii. benzylamine

5. Write the structural formula of 2-naphthol aniline dye.



Remark and sign of teacher: